



Day 0 — Prep Setup Guide (2 hours)

Repository Structure Setup

First, create your project directory structure:

```
mkdir webapp-project
cd webapp-project
mkdir data/ src/ model/ demo/
```

Python Virtual Environment Setup

Step 1: Create and Activate Virtual Environment

Create a Python virtual environment to isolate your project dependencies:[\[1\]](#) [\[2\]](#)

```
# Create virtual environment
python -m venv venv

# Activate virtual environment
# On Linux/macOS:
source venv/bin/activate

# On Windows:
# venv\Scripts\activate
```

After activation, your terminal prompt should show `(venv)` indicating the virtual environment is active.[\[3\]](#)

Step 2: Install Python Dependencies

Install all required Python packages:[\[4\]](#) [\[5\]](#)

```
pip install fastapi uvicorn transformers datasets torch tokenizers apache-log-parser requ
```

Package breakdown:

- **FastAPI:** Modern web framework for building APIs[\[6\]](#) [\[5\]](#)
- **Uvicorn:** ASGI server for running FastAPI applications[\[7\]](#) [\[4\]](#)
- **Transformers:** Hugging Face library for AI/ML models[\[8\]](#) [\[9\]](#)

- **Datasets:** Hugging Face datasets library^[10]
- **PyTorch (torch):** Deep learning framework^[8]
- **Tokenizers:** Text tokenization utilities^{[11] [12]}
- **Apache-log-parser:** Parse Apache log files^{[13] [14]}
- **Requests:** HTTP library for making API calls

Docker Installation

For Linux (Ubuntu/Debian):

```
# Update package lists
sudo apt update && sudo apt upgrade -y

# Install dependencies
sudo apt install -y ca-certificates curl gnupg

# Add Docker's official GPG key and repository
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /usr/share/keyrings/docker-archive-keyring.gpg

echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/docker-archive-keyring.gpg] https://download.docker.com/linux/ubuntu $(lsb_release -cs) stable" | sudo tee /etc/apt/sources.list.d/docker.list >/dev/null

# Update package lists and install Docker
sudo apt update
sudo apt install -y docker-ce docker-ce-cli containerd.io

# Start and enable Docker service
sudo systemctl start docker
sudo systemctl enable docker

# Verify installation
docker --version
sudo docker run hello-world
```

For Windows:

1. Download Docker Desktop from [docker.com](https://www.docker.com/products/docker-desktop)^[15]
2. Run the installer (Docker Desktop Installer.exe)
3. Enable WSL 2 during installation (recommended)^[15]
4. Restart your computer if prompted
5. Verify installation by running `docker --version` in PowerShell^[15]

OpenResty Installation

OpenResty is a web platform based on Nginx with Lua scripting capabilities.^{[16] [17]}

For Linux (Ubuntu/Debian):

```
# Install prerequisites
sudo apt-get -y install --no-install-recommends wget gnupg ca-certificates

# Import GPG key
wget -O - https://openresty.org/package/pubkey.gpg | sudo apt-key add -

# Add OpenResty repository
echo "deb http://openresty.org/package/ubuntu $(lsb_release -sc) main" | sudo tee /etc/ap

# Update package lists and install OpenResty
sudo apt-get update
sudo apt-get -y install --no-install-recommends openresty

# Start OpenResty service
sudo systemctl start openresty
sudo systemctl enable openresty

# Verify installation
openresty -v
which openresty
```

Building from Source (Alternative):

```
# Download and extract OpenResty source
wget https://openresty.org/download/openresty-1.11.2.5.tar.gz
tar -xvf openresty-1.11.2.5.tar.gz
cd openresty-1.11.2.5/

# Install build dependencies
sudo apt-get install build-essential libreadline-dev libncurses5-dev libpcre3-dev libssl-

# Configure, build, and install
./configure -j2 --with-pcre-jit --with-ipv6
make -j2
sudo make install

# Add to PATH
export PATH=/usr/local/openresty/bin:$PATH
```

Tomcat Setup and WAR File Deployment

Step 1: Install Apache Tomcat

Download and install Tomcat from tomcat.apache.org:^[18]

```
# Download Tomcat (example version 9)
wget https://downloads.apache.org/tomcat/tomcat-9/v9.0.80/bin/apache-tomcat-9.0.80.tar.gz

# Extract Tomcat
```

```
tar -xzf apache-tomcat-9.0.80.tar.gz
sudo mv apache-tomcat-9.0.80 /opt/tomcat

# Set environment variables
export CATALINA_HOME=/opt/tomcat
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64 # Adjust path as needed
```

Step 2: Deploy Sample WAR Files

There are several ways to deploy WAR files to Tomcat: [\[19\]](#) [\[18\]](#)

Method 1: Copy to webapps directory (simplest)

```
# Download a sample WAR file
wget https://tomcat.apache.org/tomcat-7.0-doc/appdev/sample/sample.war

# Copy WAR file to Tomcat webapps directory
cp sample.war $CATALINA_HOME/webapps/

# Start Tomcat
$CATALINA_HOME/bin/startup.sh
```

Method 2: Using Tomcat Manager (GUI)

1. Start Tomcat: `$CATALINA_HOME/bin/startup.sh`
2. Access Tomcat Manager at `http://localhost:8080/manager`
3. Use the "Deploy" section to upload your WAR file [\[19\]](#)

Step 3: Generate Traffic for Testing

Once your WAR file is deployed, access it to generate log traffic:

```
# Access the deployed application
curl http://localhost:8080/sample/

# Or use a web browser to navigate to:
# http://localhost:8080/sample/
```

Sample WAR Files for Testing

You can use these sample WAR files for testing: [\[20\]](#) [\[21\]](#)

1. **Tomcat Sample:** `https://tomcat.apache.org/tomcat-7.0-doc/appdev/sample/sample.war` [\[22\]](#)
2. **Hello World WAR:** Available from GitHub repositories [\[21\]](#)
3. **Custom Demo Applications:** Many tutorials provide downloadable WAR files [\[20\]](#)

Verification Steps

1. Python Environment:

```
python --version  
pip list | grep -E "(fastapi|transformers|torch)"
```

2. Docker:

```
docker --version  
docker run hello-world
```

3. OpenResty:

```
openresty -v  
curl http://localhost # Should show default OpenResty page
```

4. Tomcat:

```
curl http://localhost:8080 # Should show Tomcat welcome page  
curl http://localhost:8080/sample/ # Should show your deployed WAR app
```

Your development environment is now ready with Python dependencies, Docker, OpenResty, and Tomcat configured for web application development and log analysis. The folder structure (data/, src/, model/, demo/) provides organization for your project components.



1. <https://packaging.python.org/guides/installing-using-pip-and-virtual-environments/>
2. <https://docs.python.org/3/library/venv.html>
3. https://www.w3schools.com/python/python_virtualenv.asp
4. <https://www.geeksforgeeks.org/python/fastapi-uvicorn/>
5. <https://www.geeksforgeeks.org/python/install-fastapi-and-run-your-first-fastapi-server-on-windows/>
6. <https://alitech.io/blog/how-to-install-and-set-up-fastapi-a-step-by-step-guide/>
7. <https://pypi.org/project/uvicorn/>
8. <https://realpython.com/huggingface-transformers/>
9. <https://huggingface.co/docs/transformers/en/installation>
10. <https://www.kdnuggets.com/using-hugging-face-transformers-with-pytorch-and-tensorflow>
11. <https://github.com/huggingface/transformers/issues/20423>
12. <https://pypi.org/project/tokenizers/>
13. <https://stackoverflow.com/questions/12544510/parsing-apache-log-files>
14. <https://pypi.org/project/apachelogs/>
15. <https://docs.docker.com/desktop/setup/install/windows-install/>
16. <https://blog.openresty.com/en/ubuntu20-or-install/>
17. <https://openresty.org/en/installation.html>

18. <https://www.theserverside.com/blog/Coffee-Talk-Java-News-Stories-and-Opinions/Top-5-ways-to-deploy-a-WAR-file-to-Tomcat>
19. <https://stackoverflow.com/questions/5109112/how-to-deploy-a-war-file-in-tomcat-7>
20. <http://oak.cs.ucla.edu/refs/tomcat/index.html>
21. <https://github.com/aeimer/java-example-helloworld-war>
22. <https://middlewareadmin-pavan.blogspot.com/2021/02/deploy-sample-war-file-into-tomcat.html>
23. https://www.youtube.com/watch?v=BuGEGM_eiXY
24. <https://www.atlantic.net/dedicated-server-hosting/how-to-install-openresty-on-rocky-linux-8/>
25. <https://www.digitalocean.com/community/tutorials/how-to-install-and-use-docker-on-ubuntu-20-04>
26. https://docs.kony.com/konylibrary/konyfabric/kony_fabric_manual_install_guide/Content/Deploying_WAR_Tomcat_Auth.htm
27. <https://www.digitalocean.com/community/tutorials/how-to-use-the-openresty-web-framework-for-nginx-on-ubuntu-16-04>
28. <https://docs.docker.com/engine/install/>
29. <https://tomcat.apache.org/tomcat-9.0-doc/deployer-howto.html>
30. <https://openresty.org/en/getting-started.html>
31. <https://www.docker.com/get-started/>
32. <https://www.baeldung.com/tomcat-deploy-war>
33. <https://openresty.org/en/videos.html>
34. <https://docs.docker.com/desktop/setup/install/mac-install/>
35. <https://tomcat.apache.org/tomcat-6.0-doc/appdev/sample/>
36. <https://www.linode.com/docs/guides/using-openresty/>
37. <https://docs.n8n.io/hosting/installation/docker/>
38. <https://www.geeksforgeeks.org/python/python-virtual-environment/>
39. <https://code.visualstudio.com/docs/python/environments>
40. <https://fastapi.tiangolo.com/virtual-environments/>
41. <https://code.visualstudio.com/docs/python/tutorial-fastapi>
42. <https://docs.parasoft.com/display/SOA9105/Apache+Tomcat+Example+Deployment>
43. <https://www.freecodecamp.org/news/how-to-setup-virtual-environments-in-python/>
44. <https://www.middlewareinventory.com/blog/sample-web-application-war-file-download/>
45. <https://fastapi.tiangolo.com/tutorial/>
46. <https://github.com/Azure-Samples/tomcat10-jakartaee9>
47. <https://fastapi.tiangolo.com/deployment/manually/>
48. <https://www.piwheels.org/project/apache-log-parser/>
49. <https://dev.to/willthefarmer/apache-log-parser-and-data-normalization-farm-built-3ahl>
50. <https://pypi.org/project/transformers/>
51. <https://www.theserverside.com/video/Step-by-step-Maven-Tomcat-WAR-file-deploy-example>
52. <https://opensource.com/article/20/7/python-lars>
53. https://pytorch.org/hub/huggingface_pytorch-transformers/

54. <https://pypi.org/project/apache-log-parser/>

55. https://colab.research.google.com/drive/1hmVMdHsxcauVjloOWrpM0a_ISFKVjyCX